
Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models now answer hard scientific questions, yet they still hallucinate and follow broken lines of reasoning. Multi-agent debate offers one way to push back on this. Several models work through a question before committing to an answer. The reasoning usually gets sharper, and errors get caught. The real problem is how these debates settle. Most systems pick a winner by majority vote or by which agent sounds most confident. A confident majority that is simply wrong can pull a correct minority agent off its answer, and the group lands on the wrong conclusion. Current methods never check whether an agent’s claims hold up against outside scientific evidence before letting that agent sway the others. This work changes how much influence each agent carries while the debate runs. Every agent response is split into small factual claims. For each claim, relevant scientific literature is retrieved and checked to see whether the evidence backs it or contradicts it. Each agent carries a trust score that moves up or down as these checks accumulate over the debate rounds. Unsupported claims lose influence, so a correct minority agent keeps its standing even when outnumbered. The final answer comes from trust-weighted aggregation, not a raw count of votes. The framework is expected to reduce hallucinations and keep reasoning tied to verifiable sources. It may also reduce premature agreement on a wrong answer while operating without model fine-tuning.

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Publication List

The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1. Leveraging Machine Learning in Balancing Inventories and Analyzing Market Trends: A Comprehensive Systematic Review (Artificial Intelligence Review, Under processing)

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

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Chapter 1

Introduction

The chapter introduces the study. It presents the background and problem, the motivation behind the work, the objectives, a short summary of the methodology, the expected outcome, and the structure of the rest of the report.

1.1 Project Overview

Large language models now handle many reasoning tasks, from scientific question answering to tutoring and decision support. A single model can still be confidently wrong. It may state an incorrect answer in fluent and assured language, which makes the mistake hard to catch. Multi-agent debate was proposed to reduce this problem. Several agents answer the same question, read each other’s reasoning, and revise their positions over a few rounds before a final answer is selected [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes the reasoning inside a debate easier to inspect. The real difficulty is the final step, where the separate answers are combined. Most debate systems settle disagreements by majority vote. Each agent receives the same weight, so the most common answer wins. This rule treats popularity as a stand-in for correctness. In scientific reasoning, a better answer should be backed by evidence rather than repeated by more agents. The study examines a trust-calibrated multi-agent deliberation framework that adjusts each agent’s influence during the debate based on how well its claims match retrieved external evidence. The final answer then depends on evidence-grounded trust rather than a raw headcount.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own, and that the correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases [3]. Work on model behaviour

points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [4]. When a confident majority pushes a correct minority agent to give up its answer, majority voting does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Each existing method attacks only part of the problem. iMAD decides when a debate is worth running at all [5]. DebUnc scales agent influence by self-reported confidence [6]. Mixture-of-Agents folds all outputs together through static aggregation [7]. None of these ties an agent’s influence to whether its claims survive a check against external evidence while the debate is still running. That connection is exactly what this study adds.

1.3 Objectives

The main objective is to reduce sycophantic consensus in multi-agent LLM debate. Plain majority voting gets replaced with evidence-grounded trust calibration. Reaching that goal needs a few things. The first is a clear picture of how a confident wrong majority pressures a correct minority agent into changing its answer [3, 8]. The second is a bounded trust score. It rises when an agent’s claims are supported by external evidence and falls when they are contradicted, and it is applied before the final answer is fixed. The study also compares the method against standard majority voting and other debate and aggregation baselines [9, 10], using answer accuracy together with consensus-collapse, minority-preservation, and evidence-calibration metrics.

1.4 Methodology

The study follows an experimental design. A baseline multi-agent debate system is built first. Several heterogeneous LLM agents answer the same scientific question, read each other’s reasoning, and revise their positions across rounds [1]. The baseline keeps majority voting so the usual failure mode can be observed and measured [9]. Each agent’s reasoning is then split into smaller factual claims. The claims are checked against scientific evidence retrieved from external sources such as academic papers and reference databases [11]. Contradictions between a claim and the retrieved evidence are detected and scored [12]. This builds on the idea that external tools can verify and correct model output [13]. The verdicts feed into trust updates, following the line of work that uses per-agent confidence to steer debate [6]. Supported claims raise an agent’s trust. Contradicted claims lower it. Uncertain claims are handled conservatively. The trust score stays bounded, so no agent gains unlimited influence or drops out of the debate completely. The final answer comes from trust-weighted aggregation rather than a simple count. The framework is tested on scientific reasoning datasets such as GPQA and MMLU-Pro. The evaluation looks at three things: whether the mechanism lowers consensus collapse, whether correct minority answers survive, and whether overall accuracy holds, all without any model fine-tuning.

Figure 1.1 shows the end-to-end flow of the framework.

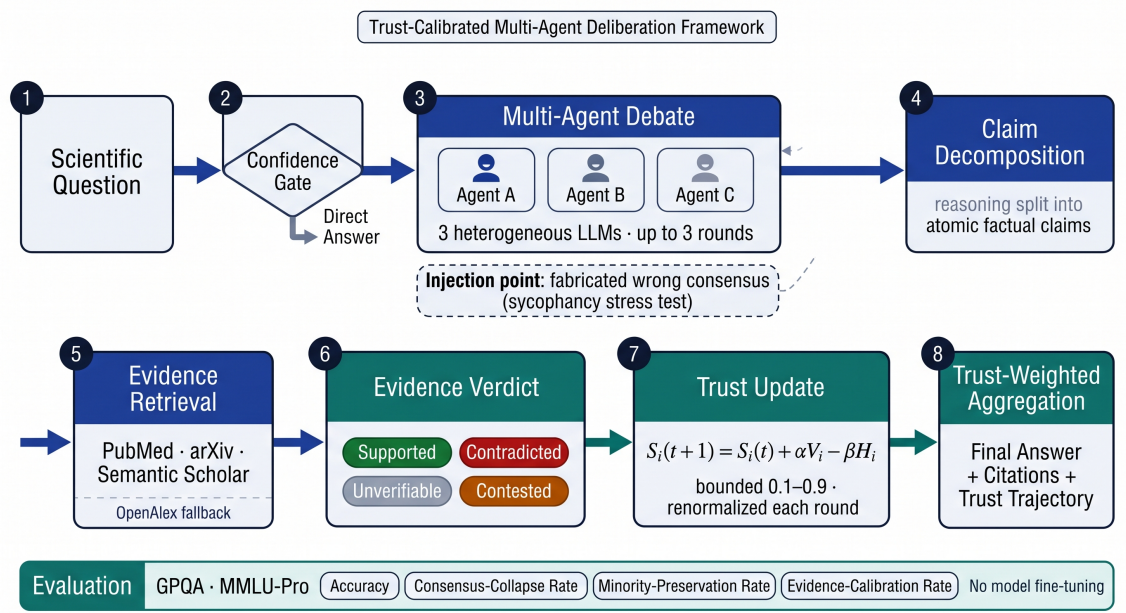


Figure 1.1: End-to-end flow of the trust-calibrated multi-agent deliberation framework: debate, claim-level evidence verification, per-round trust updates, and trust-weighted aggregation.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated multi-agent scientific deliberation. Retrieved evidence updates agent influence while the debate runs, and the final answer rests on evidence-weighted trust instead of a majority vote. The study also produces an evaluation protocol. It detects sycophantic consensus [3], tracks how often correct minority answers survive [8], and measures the method against established baselines. The hope is that dynamic trust calibration makes multi-agent debate more dependable on scientific questions. Where the method fails to remove every wrong answer, it should still make the decision more transparent by recording which agents the evidence supported and which it contradicted.

1.6 Organization of the Report

The rest of the report is arranged as follows. Chapter 2 covers the background, the related work on multi-agent debate and sycophancy, and the gap addressed by the study. Chapter 3 presents the requirement analysis, the detailed methodology, the system design, and the project plan. Chapter 4 describes the implementation setup, the testing and evaluation procedure, and the results with discussion. Chapter 5 discusses the relevant standards, the

impact of the work, sustainability, and the complex engineering problem analysis. Chapter 6 concludes the report, notes its limitations, and gives recommendations for future work.

Chapter 2

Background

The chapter introduces the concepts needed to understand the system described in the report and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by the study.

2.1 Preliminaries

Large language models (LLMs) generate answers one token at a time, but their outputs can still contain unsupported or incorrect reasoning. Fluent wording does not guarantee that an answer is correct [14]. Chain-of-thought prompting asks a model to expose intermediate reasoning steps before giving a final answer, which makes the reasoning easier to inspect and gives later components more information to evaluate [2]. These steps are useful evidence about how an answer was formed, but they should not be treated as proof of correctness by themselves. Multi-agent debate takes this further. Several agents solve the same problem on their own, then see the other answers and explanations, discuss the differences, and revise their positions across a fixed number of rounds. Only after the discussion is a final answer chosen [1]. Independent attempts help because different agents expose different errors, and one agent can question another agent’s reasoning. How the answers get combined still matters. Majority voting is the common choice: every agent carries the same weight and the most frequent answer wins. The vote then strengthens a repeated mistake instead of correcting it [10]. Sycophantic consensus describes the case where an agent copies a wrong position or gives up a correct one simply because the wrong position has more support [3]. Confidence can come from the agent itself or from its token distribution, but neither source verifies the underlying claim [6]. DebUnc makes this point by comparing deployable uncertainty measures against a diagnostic Ground Truth measure that already knows the answer [6]. The framework instead builds trust from retrieved scientific evidence.

2.2 Literature Review

The review sorts the literature into five areas: debate construction, external verification, sycophancy, uncertainty, and debate evaluation. The main comparison sits between methods that decide when to debate, methods that change agent interaction, and the method that changes agent influence using retrieved evidence. Surveys of LLM-based multi-agent systems map the design space across workflow, infrastructure, and open challenges [15]. Recent reasoning-oriented LLMs show that better reasoning needs more than a bigger model. Research has moved toward structured reasoning, multi-agent collaboration, and external verification to make generated responses more reliable [14]. Multi-agent debate is among the most studied of these directions, because agents can challenge each other’s reasoning before a final answer is produced [1]. The following review goes through representative studies in these areas and points to the research gap the framework addresses.

2.2.1 Similar Applications

The system is not a conventional web or mobile application. Instead, it is an experimental deliberation layer that can be integrated into scientific question-answering systems, research assistants, evidence-based decision-support tools, or other applications that require reliable reasoning. Rather than relying on a single model response, these systems generate multiple candidate solutions, allow the candidates to interact, and aggregate the results before returning a final answer to the user. Du et al. demonstrated that structured debate among multiple agents can improve factuality and reasoning compared with a single-agent approach [1]. Wang et al. further showed that combining the outputs of several models through layered aggregation can improve overall model capability without requiring explicit debate [7]. Existing multi-agent applications differ mainly in how they coordinate agents and combine their outputs. Some systems focus on structured debate to encourage reasoning refinement between agents [1]. Other systems organize multiple agents into hierarchical aggregation pipelines where later models refine or combine earlier responses [7]. Zhou et al. described a self-organizing setup where the task itself decides which reasoning agents take part, instead of a fixed workflow [16]. Kushwaha and Madhavan built a retrieval-augmented system that spots contradictions between retrieved documents and ranks evidence by source credibility before producing a final response [12]. Both directions show that spreading reasoning across specialized agents helps robustness and factual consistency. Their aggregation stays fixed once the discussion starts. The proposed framework keeps the same goal, reliable answers, but lets agent influence change during the discussion. Evidence verification results are recorded after every round, and each agent’s trust score is updated from claim-level scientific evidence. That way a well-supported minority opinion can survive, and an unsupported majority consensus carries less weight.

2.2.2 Related Research

Wei et al. (2022) showed that chain-of-thought prompting improves reasoning in large language models [2]. Du et al. (2024) built on that with multi-agent debate, where agents answer a question, look at peer reasoning, and revise their positions [1]. CRITIC takes a different route [13]. It uses external tools to check and improve a model’s response, so claims can be verified, but only one model works on the question at a time. Mixture-of-Agents is layered instead of conversational [7]. Several proposer models produce answers and a later stage combines them, which shows what multiple LLMs can do, yet the combination is static and carries no evidence-based trust value. iMAD attacks cost rather than correctness [5]. Fan, Yoon, and Ji extract 41 features from a structured self-critique and let a classifier decide whether a debate should start at all. They report token savings of up to 92 percent against GroupDebate and accuracy gains of up to 13.5 percentage points over a single-agent baseline. Once the debate begins, agent influence stays unchanged. Pitre, Ramakrishnan, and Wang studied sycophancy inside debate [3]. In their experiments the correct answer was present but ignored in more than 20 percent of wrong-answer cases. CONSENSAGENT detects stalling and answer copying, then rewrites the task prompt before the next debate stage. Measured sycophancy drops, but the final aggregation still leans on self-reported confidence and consistency. DebUnc instead shares uncertainty between agents [6]. Uncertainty reaches peers through prompts or attention scaling, yet the deployable measures give small gains. A diagnostic Ground Truth signal, which knows the correct answer, gives a much larger gain, and the gap between the two exposes the real bottleneck: the quality of the trust signal. Kushwaha and Madhavan described an eight-agent MAS-RAG system that detects contradictions among retrieved documents and ranks evidence by credibility [12]. Source credibility here is fixed by a human curator and never updated while deliberation runs. Other studies look at the social side of debate. Cau et al. found that selective persuasion can steer opinion convergence [17]. Yeow et al. compared decision architectures and found that feedback between agents can spread early errors [18]. Wang et al. showed that a single LLM with strong in-context examples can sometimes match multi-agent discussion, and they list judge errors and peer conformity as weak points [10]. Choi, Zhu, and Li separated debate communication from voting and argued that majority voting explains much of the usual gain [9]. FREE-MAD goes further and drops final consensus altogether, using anti-conformity prompts and trajectory scoring [19]. ReSo selects agents through reward-driven dynamic task graphs [16]. These works tune coordination and agent selection, but none verifies scientific claims before changing agent influence. Sycophancy also appears at the model level. Ibrahim et al. reported that tuning models to sound warm and agreeable reduces factual accuracy and increases sycophancy [4]. Chen et al. attacked the same problem from the training side with self-augmented preference alignment [20]. Pitre et al. proposed process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility, useful when ground truth is unavailable [21]. Final agreement alone, their

work suggests, is not enough to judge a debate. Across all of these studies, no method continuously updates an agent’s influence from claim-level support in retrieved scientific evidence. That is the gap the framework closes with a bounded trust score.

2.3 Gap Analysis

The reviewed literature reveals three remaining needs in multi-agent debate systems. First, a system should determine whether the additional computational cost of the debate is justified. Fan et al. addressed this through iMAD by predicting whether a debate is necessary before it begins [5]. Second, a system should detect unsupported agreement caused by conformity or sycophancy. CONSENSAGENT mitigates this problem through prompt optimization during debate [3]. Third, a system should determine which agents deserve greater influence when disagreement remains. DebUnc introduces uncertainty-based weighting, but its trust signals rely on internal model uncertainty rather than independently verifiable evidence [6]. Previous studies also show that agreement does not necessarily indicate correctness. Cau et al. found that selective persuasion influences opinion convergence [17]. Wang et al. identified peer conformity as a limitation of multi-agent debate [10]. Existing retrieval-based systems verify retrieved information or improve aggregation, but they do not continuously update an agent’s influence according to verified scientific evidence during debate [13]. Dynamic coordination methods also focus on agent selection rather than evidence-driven trust updates [16]. The remaining gap is therefore an evidence-grounded mechanism that continuously updates agent influence during an ongoing debate. The proposed framework addresses this gap by decomposing an agent’s response into factual claims, retrieving relevant scientific passages, and classifying each claim as supported, contradicted, or unverifiable. These verification results update a bounded trust score after every debate round, allowing well-supported minority agents to retain influence while reducing the impact of unsupported majority consensus. The framework is evaluated using consensus-collapse rate, minority-preservation rate, and evidence-calibration rate alongside conventional answer accuracy. Retrieval sparsity, imperfect claim decomposition, and correlated reasoning errors between agents remain limitations of the proposed approach.

2.4 Summary

The chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The reviewed studies cover debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. They do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

This chapter defines the system requirements, presents the detailed methodology and the design decisions behind the framework, and lays out the project plan with the task allocation for the team. The first section records the functional and nonfunctional requirements together with the context diagram, the level-one data flow diagram, and the interface design.

3.1 Requirement Analysis

Requirement analysis fixes what the framework must do before any code is written. The system is an experimental deliberation layer rather than a conventional application, so the requirements are framed around the research pipeline. A question enters, agents debate it over a fixed number of rounds, the claims are checked against external evidence, and a trust-weighted answer comes out. Multi-agent debate was chosen because it has been shown to improve factuality over single-model answers [1]. The trust mechanism exists because a confident wrong majority can push a correct minority agent into abandoning its answer [3]. The subsections below define the requirements, the system boundary, the data flows, and the interface through which the framework is operated.

3.1.1 Functional and Nonfunctional Requirements

The functional requirements describe the services the framework must provide. The system accepts a scientific question from a user or the evaluation harness and keeps the ground truth hidden from all agents. For every question the confidence gate decides whether a debate is needed, sends easy questions to a direct answer, and routes the remaining questions into the debate pipeline. The orchestration process runs three heterogeneous LLM agents over at most three revision rounds, with a retry cap of three, and sequences the rounds with a state machine. Each agent’s reasoning is split into atomic factual claims tagged to their source agent, with a fallback extraction pass when structured tagging fails. The retrieval subsystem searches each agent’s claims in a different database, PubMed for the first agent, arXiv for the second, and Semantic Scholar for the third, with

OpenAlex as a fallback when a source is unavailable. Retrieved passages are reranked with a cross-encoder and every claim receives a verdict of supported, contradicted, unverifiable, or contested. After every round the trust score of each agent is updated with $S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i$, then softmax is applied, the values are clamped between 0.1 and 0.9, and the vector is renormalized so that no agent is silenced or dominant. The system also supports an injection point where a fabricated wrong consensus can be inserted between rounds, so that sycophantic collapse can be measured under controlled pressure. The final answer is selected by weighted aggregation of trust and position at the last round rather than by majority vote, and the result package contains the answer, the supporting citations, the full trust trajectory, and each agent’s reasoning trace. The evaluation harness computes answer accuracy, consensus-collapse rate, minority-preservation rate, and evidence-calibration rate, and compares the framework with majority voting and with the MoA [7], iMAD [5], and DebUnc [6] baselines. Finally, the system recovers unparseable model output with a fallback extraction pass, retries a timeout once before marking the round inconclusive, and falls back to cached evidence or OpenAlex when a retrieval API is down.

The nonfunctional requirements describe the qualities of the system rather than its functions. Latency is the first constraint. The debate must respect the round cap of three and the retry cap of three, so latency stays bounded on the target hardware. A failure in one component does not block the whole debate. Every failure path degrades locally and keeps the pipeline moving. Portability matters too. Closed APIs and locally served models both work through the OpenAI-compatible vLLM interface. The pipeline runs unchanged on the development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B) and on the final models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B). Inference fits within a single A100 80GB rental at about USD 0.68 to 1.50 per hour, with no fine-tuning. Every run is seeded and every intermediate result is logged, so any experiment can be repeated and checked. The output records which agent was supported or contradicted by the evidence, leaving a decision trail that can be audited. The ground truth stays hidden from the agents, and the final output cites the evidence behind the answer.

3.1.2 Context Diagram

The context diagram in Figure 3.1 shows the system boundary and the external actors it exchanges data with. The framework sits in the middle of the diagram. It receives questions from the user or the evaluation harness, retrieves scientific passages from three external databases, and calls off-the-shelf language models through the vLLM server running on a rented A100 GPU. The evaluation harness receives the scores produced by the framework, and the user receives the final result package with the answer, the evidence, and the trust trajectory.

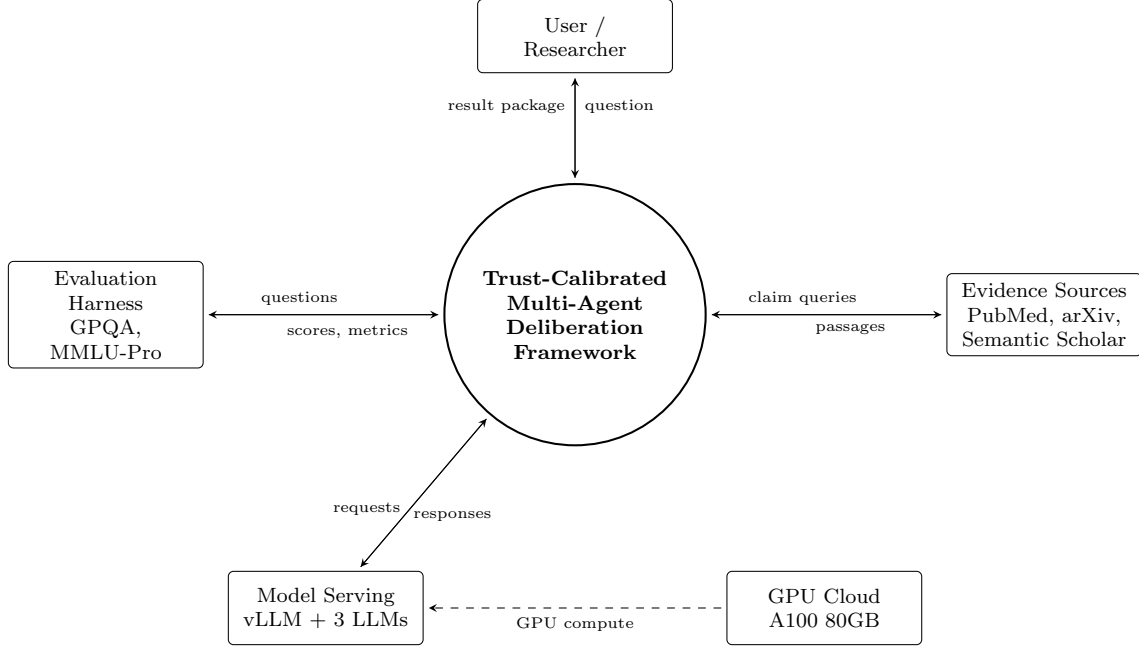


Figure 3.1: Context diagram of the trust-calibrated multi-agent deliberation framework.

3.1.3 Data Flow Diagram Level 1

The level-one data flow diagram in Figure 3.2 decomposes the framework into seven processes. A question enters through the intake and gate process, which decides between a direct answer and a debate. The orchestration process runs the revision rounds, sends inference requests to the external models, and reads and writes the debate state. Each position is decomposed into atomic claims and verified against the external sources. The verdicts drive the trust update, which writes the new trust values back into the debate state. At the last round the weighted aggregation process selects the answer, and the packaging process logs the result and returns it to the user. Three data stores hold the debate state, the evidence and verdict log, and the results log.

The diagram reads from left to right and top to bottom. The evaluation harness submits a question, the confidence gate decides whether the debate is worth running, and the orchestration process asks the model server for the agent positions. Each position is decomposed into atomic claims, the retrieval process verifies them against the external sources, and the verdicts are stored in the evidence log before they drive the trust update. The updated trust is written back into the debate state, so the next round starts from the corrected influence values. At the last round the aggregation process selects the answer, the packaging process logs it in the results log, and the result package with the answer and the citations returns to the user.

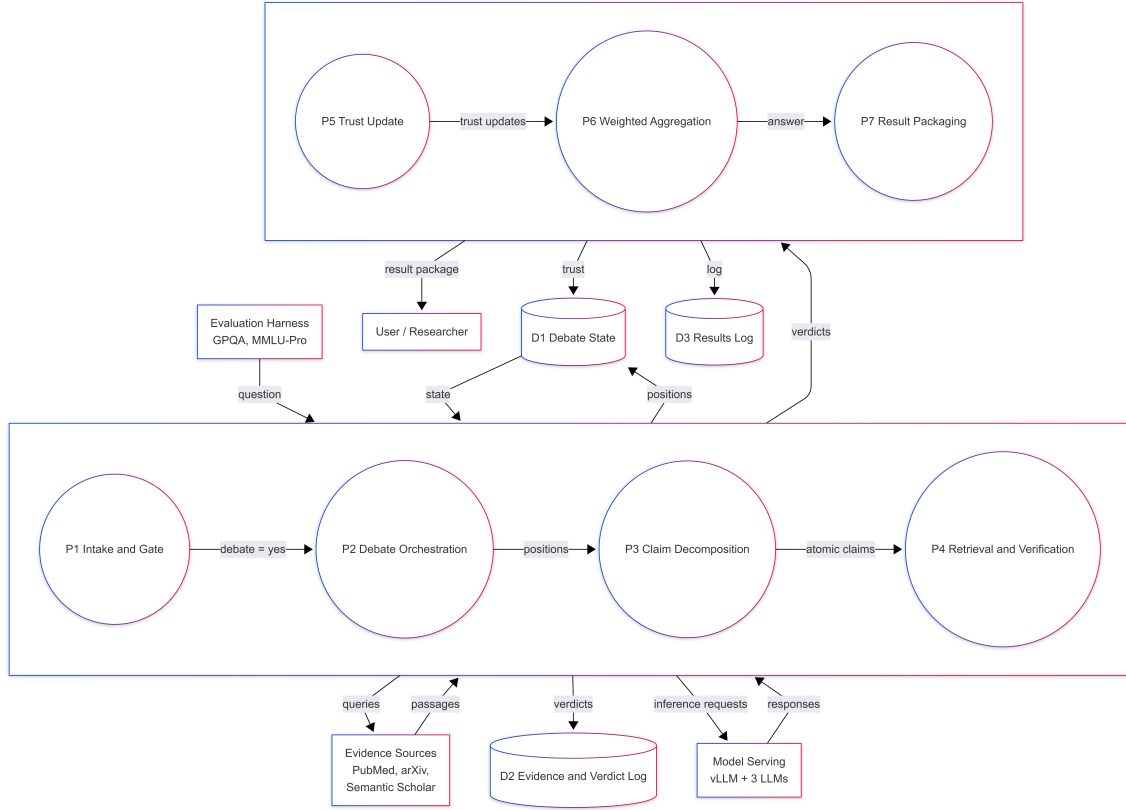


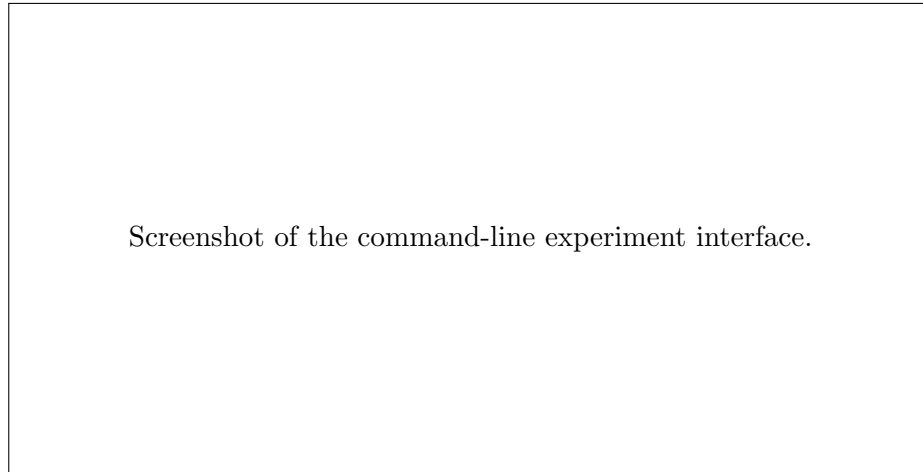
Figure 3.2: Level-one data flow diagram of the trust-calibrated multi-agent deliberation framework.

3.1.4 UI Design

The framework is operated through a command-line interface and a small web dashboard. The command-line interface starts experiments, points the pipeline at the evaluation datasets, and writes the result of every run to a JSON package. It is the primary interface because the system runs in batches on a rented GPU, where a graphical interface adds little value. The web dashboard is used afterwards to inspect single debates. It shows the agents as cards with their current positions, a chart of the trust trajectory across rounds, and the verdict of every claim with the passage that supports or contradicts it. This is where the transparency of the framework becomes visible, because a user can trace exactly why an agent lost influence. Figure 3.3 and Figure 3.4 show the intended layout of both interfaces. The screenshots will be replaced with the real output of the system once the implementation is complete.

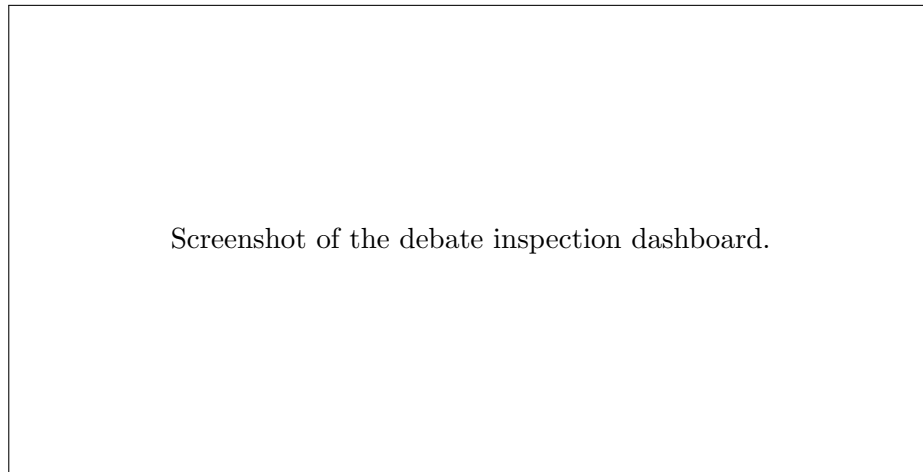
3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.



Screenshot of the command-line experiment interface.

Figure 3.3: Command-line interface for running experiments and writing JSON result packages.



Screenshot of the debate inspection dashboard.

Figure 3.4: Web dashboard showing agent positions, the trust trajectory, and per-claim evidence verdicts.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only include standards related to the system. This list is not exhaustive. For each standard, discuss the alternatives, their advantages and disadvantages, and the reason for selecting it.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only include constraints related to the design. This list is not exhaustive.

5.2.1 Economic Constraint**5.2.2 Environmental Constraint****5.2.3 Ethical Constraint****5.2.4 Health and Safety Constraint****5.2.5 Social Constraint****5.2.6 Political Constraint****5.2.7 Sustainability****5.3 Cost Analysis**

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem**5.4.1 Complex Problem Solving**

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowl- edge	P2 Range of Con- flicting Require- ments	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stake- holder Involve- ment	P7 Inter- dependence
✓	✓					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

Table 5.2: Mapping with complex engineering activities.

A1 Range of re- sources	A2 Level of Interac- tion	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓	✓			

5.5 Summary

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

References

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